

$$\Rightarrow A + \bar{A}C = (A + \bar{A}) \cdot (A + C)$$

$$= (1) \cdot (A + C) = A + C$$

$$\Rightarrow A \oplus \bar{A}C = (A \oplus \bar{A}) \cdot (A \oplus \bar{A}C)$$

$$= (\bar{A} + A) \cdot (\bar{A}C + A)$$

$$= \bar{A}C + AC$$

$$\Rightarrow A \oplus \bar{A}C = A(\bar{A}C) + (\bar{A}C)A$$

$$= A(\bar{A}C) + \bar{A}C$$

$$A = A + \bar{A}C + \bar{A}C$$

$$= A(1 + \bar{C}) + \bar{A}C = A + \bar{A}C$$

$$= A + C$$

$$\boxed{(\star) X + \bar{X}Y = X + Y}$$

END OF
DAY-4

$$AB + \bar{A}B + A\bar{B} + \bar{A}\bar{B} = (A + \bar{A})(B + \bar{B})$$

$$AB + (\bar{A}B + A\bar{B} + \bar{A}\bar{B}) =$$

$$AB + A =$$

$$(A + \bar{A}) \cdot (B + \bar{B}) = (A + \bar{A}) \cdot 1 = A + \bar{A} = 1$$

$$(A + \bar{A})(B + \bar{B}) = 1 \cdot 1 = 1$$

DAY-5

(04-06-26)

$$\Rightarrow X \oplus Y \oplus XY = X \oplus [Y(1 \oplus X)]$$

$$= X \oplus Y\bar{X} = X + Y$$

$$(Q) \textcircled{1} (A+B)(\bar{A}+\bar{B}) = (B+A)(\bar{B}+\bar{A}) = B + A\bar{A}$$

$$= B$$

$$\textcircled{2} (A+C)(A+D) = (A+B+C)(A+B+D)$$

$$= A+B+C$$

$$\textcircled{3} \bar{A} + AB = \bar{A} + (\bar{A}) \cdot B = \bar{A} + B$$

$$\textcircled{4} \bar{A} + AB = \bar{A} + (\bar{A}) \cdot B = \bar{A} + B = \bar{A}B$$

$$\textcircled{5} PQ + \bar{P}Q RS = PQ + RS$$

$$\textcircled{6} \underbrace{AB + \bar{A}B + A\bar{B} + \bar{A}\bar{B}} = A(B + \bar{B}) + \bar{A}(B + \bar{B})$$

$$= A + \bar{A} = 1$$

$$(\star) AB + BC + CA = \bar{A}C + AB \quad [\text{Consensus Theorem}]$$

Proof:- $AB + \bar{A}C + BC = AB + \bar{A}C + AB\bar{C} + A\bar{B}C$

$$= AB(1 + \bar{C}) + \bar{A}C(1 + B)$$

[SOP]

$$= AB + \bar{A}C$$

$$\Rightarrow \frac{\overline{A}B + \overline{B}C + \overline{C}A}{x} = \overline{A}B + \overline{C}A \quad (\text{Find one term \& its conjugate, leave 3rd term}).$$

$$\Rightarrow \frac{AB + \overline{B}C + \overline{C}A}{x} = \overline{B}C + \overline{C}A$$

$$\Rightarrow \frac{\overline{A}B + \overline{B}C + \overline{C}A}{x} = \text{Not Applicable}$$

$$\Rightarrow \frac{\overline{A}B + \overline{B}C + \overline{C}A}{x} = \overline{A}\overline{B} + \overline{B}\overline{C}$$

$$\Rightarrow (A+B)(B+C)(C+A) = (A+B)(\overline{A}+C) \quad (\text{Pos})$$

$$\Rightarrow \text{Proof: } (A+B)(\overline{A}+C)(B+C+A\overline{A})$$

$$= (A+B)(\overline{A}+C)(B+C+A)$$

$$= (A+B)(A+B+C)(\overline{A}+C)(\overline{A}+B+C)$$

$$= (A+B)(\overline{A}+C)(\overline{A}+B+C)$$

$$= \underline{\underline{(A+B)(\overline{A}+C)}}$$

$$\Rightarrow (A+B)(B+C)(C+A) = (A+B)(B+C)$$

$$\Rightarrow (A+B)(B+C)(\overline{C}+A) = (B+C)(\overline{C}+A)$$

$$\Rightarrow (\overline{A}+\overline{B})(B+\overline{C})(\overline{C}+\overline{A}) = (\overline{A}+\overline{B})(B+\overline{C})$$

$$\Rightarrow \frac{(\overline{A}+B)(\overline{B}+C)(C+A)}{x} = \underline{\underline{(\overline{B}+C)(C+A)}}$$

$$\Rightarrow \frac{\overline{A}B + \overline{A}C + \overline{B}C}{x} = \underline{\underline{\overline{A}B + \overline{B}C}}$$

$$\Rightarrow \frac{\overline{B}C + \overline{A}B + \overline{A}C}{x} = \underline{\underline{\overline{B}C + \overline{A}B}}$$

(a) Simplify

$$(i) \overline{A}BC + \overline{A}BC + \overline{A}BC$$

$$= \overline{A}BC + \overline{A}C[B+B] = \overline{A}BC + \overline{A}C$$

$$= C[\overline{A} + \overline{A}B]$$

$$= C \cdot \overline{A} + B = \underline{\underline{\overline{A}BC}}$$

$$(ii) \overline{A}B + \overline{A}B\overline{C} + \overline{A}BDE + \overline{A}B\overline{C}DE$$

$$= \overline{A}B[1 + \overline{C} + DE + \overline{C}DE]$$

$$= \underline{\underline{\overline{A}B}}$$

$$(iii) AB + \overline{A}C + BCDEF = AB + \overline{A}C + BCDEF(A+\overline{A})$$

$$= AB + \overline{A}C + ABCDEF + \overline{A}BCDEF$$

$$= AB(1 + CDEF) + \overline{A}C(1 + BDEF)$$

$$= \underline{\underline{AB + \overline{A}C}}$$

$$\begin{aligned}
 \text{(iv) } ABC + BED + ABD &= ABC + BED + ABD (C+E) \\
 &= ABC + BED + ABCD + ABED \\
 &= ABC(1+D) + BED(1+A) \\
 &= \underline{ABC + BED}
 \end{aligned}$$

⇒ De Morgan's Theorem :-

$$\Rightarrow \overline{AB} = \overline{A} + \overline{B}$$

$$\Rightarrow \overline{A+B} = \overline{A} \cdot \overline{B}$$

$$\Rightarrow \overline{ABC} = \overline{A+B+C} = [\overline{A+B}] \cdot \overline{C} = (\overline{A} \cdot \overline{B}) \cdot \overline{C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

$$\Rightarrow \overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

$$\Rightarrow \overline{ABCD} = \overline{A+B+C+D} = \overline{A+B+C} \cdot \overline{D} = (\overline{A} \cdot \overline{B} \cdot \overline{C}) \cdot \overline{D} = \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D}$$

$$\Rightarrow \overline{A+B+C+D} = \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D}$$

⇒ Simplify :-

$$① \overline{A \cdot B \cdot C} = \overline{A+B+C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

$$② \overline{(A+B) \cdot (C+D)} = \overline{A+B+C+D} = \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D}$$

$$③ \overline{A \cdot B \cdot C \cdot D} = \overline{A+B+C+D} = \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D}$$

$$④ \overline{(A+B+C+D)} + \overline{ABCD} = \overline{A+B+C+D} \cdot \overline{ABCD} = \overline{A} \cdot \overline{B} \cdot \overline{C} \cdot \overline{D} + \overline{A+B+C+D}$$

$$\Rightarrow \overline{A} \overline{B} C + \overline{A} \overline{B} \overline{C} + \overline{A} B \overline{C} D$$

$$= \overline{A} \overline{B} C + \overline{A} \overline{B} \overline{C} + \overline{A} B \overline{C} D$$

$$= \overline{A} \overline{B} (C + \overline{C} D) = \overline{A} \overline{B} (C + D)$$

$$\Rightarrow \overline{A} \overline{B} C + \overline{A} B \overline{C} + \overline{A} B C$$

$$= \overline{B} C [A + \overline{A}] + \overline{A} B C$$

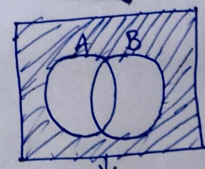
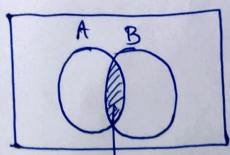
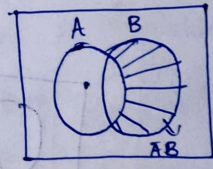
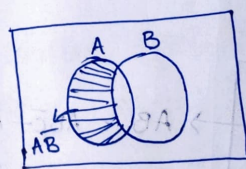
$$= \overline{B} C + \overline{A} B C = C [\overline{B} + \overline{A} B]$$

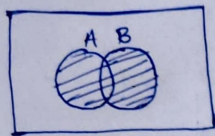
$$= C [\overline{B} + \overline{A}]$$

$$= C [\overline{AB}]$$

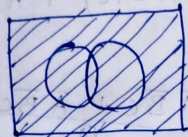
⇒ Venn diagrams :-

2-Variable :-





$$(A+B)$$

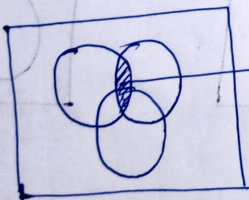
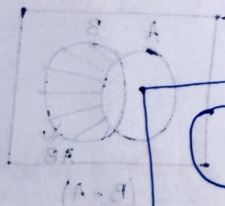
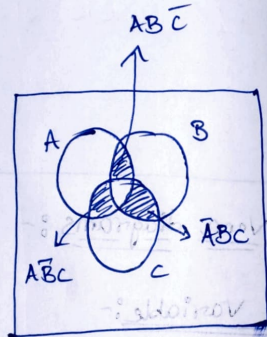
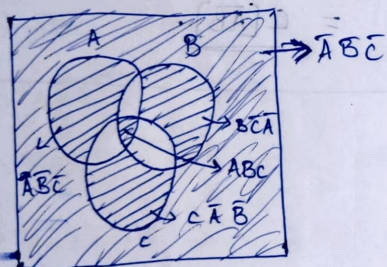


$$(1)$$

$$\overline{A}B + A\overline{B} + AB + \overline{A}\overline{B} = (A+B) + (\overline{A+B})$$

$$= 1$$

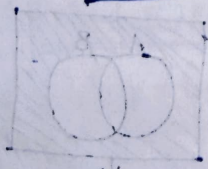
3-Variable:-



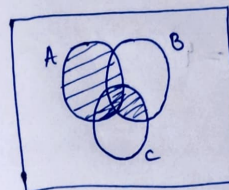
$$AB = A\overline{B}\overline{C} + AB\overline{C} + ABC$$

$$= AB(\overline{C} + C)$$

$$= AB$$



(Q) Shaded area Represents



$$= A + \overline{B}C\overline{A}$$

$$= \underline{A + \overline{B}C}$$

Distinct logical expression:-

(Same) $\rightarrow$ A
 $A+AB$
 $= A(1+B)$
 $= A$

$\leftarrow$ All 3 different
 $A+B$
 (Distinct logical EXP., Different Truth Tables)

why A and A+AB are same?

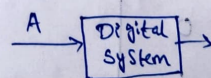
A	B	A	A+AB
0	0	0	0
0	1	0	0
1	0	1	1
1	1	1	1

(Same Truth Table)

Total no. of logical expressions with variables:-

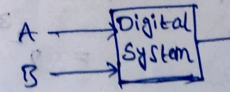
(Q) Suppose you are given 1 variable. How many "distinct logic expressions" you can make?

A:



$$\frac{A}{A} = 1$$

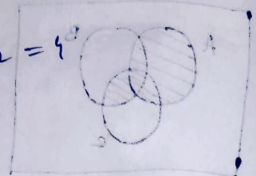
(*) For 2 variables,



⇒ Variable A and B can have $2^2 = 4$ Possible combinations.

A	B
0	0
0	1
1	0
1	1

$$2 \times 2 = 4$$



With respect to each combination, 2 outputs ["0" or "1"] are possible and each one has 2

⇒ 4 combinations outputs ⇒ $2^4 = 16$ Distinct logic expressions.

(*) For 3 Variables

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

⇒ 8 combinations & 2 o/p each = $2^3 = 8$

(*) For n Variables, 2^n distinct logical / Boolean expression is there.

(*) Literal:- Each variable in a logical expression.

• Eg:- (i) $AB + CD + BD \rightarrow L.C = 6$

(ii) $AB + \bar{A}B + DE + FG \rightarrow L.C = 8$

(iii) $(A+B+C)(D+E+F) \rightarrow L.C = 6$

(*) Taking dual of logic expression:-

(i) "+" $\leftrightarrow$ "."

(ii) "0" $\leftrightarrow$ "1"

(iii) Don't complement the literals.

Eg:- (i) $f = 0$ (ii) $f = 1$

$f^D = 1$

$f^D = 0$

(iii) $f = A+B$

(iv) $f = AB$

$f^D = \bar{A} \cdot \bar{B} + 1$ (v) $f^D = A+B$

(vi) $f = (A+B) \cdot (B+C) + CD$

$f^D = [AB + (B+C) \cdot (C+D)]$

(vii) $f = AB + BC + CA$

$f^D = (A+B) \cdot (B+C) \cdot (C+A)$

$$= [AB + AC + B + BC] [A + C]$$
~~$$= AB + AC + AB + AC + AB + AC + B + BC + B + BC$$~~

$$= [B(A + 1 + C) + AC] [A + C]$$

$$= [B + AC] [A + C] = AB + BC + CA$$

$$\boxed{f^D = f} \quad (\text{Self-dual function}).$$

$$\boxed{(\star) (f^D)^D = f}$$

(*) Taking complement of a fn. "f" using duality :-

$\Rightarrow$ Complement of $f(f^c)$:-

(*) $f \xrightarrow[\text{dual}]{\text{Take dual}} f^D \xrightarrow[\text{each literal individually}]{\text{complement}} f^c$

(a) Take complement of $f = (a + \bar{b}c)d + \bar{e}f + \bar{g}$

$$\underline{A.} \quad f^D = [\bar{a} \cdot \bar{b} + \bar{c}] + d \cdot \bar{e} + f \cdot \bar{g}$$

$$f^c = [\bar{a}b + \bar{c}] + \bar{d} \cdot e + \bar{f} \cdot g$$

(a) Take complement of $f = AB + BC + CA$
(Self-dual fn)

$$\underline{A.} \quad f^D = (A + B) \cdot (B + C) \cdot (C + A)$$

$$= [AB + AC + B + BC] \cdot (C + A)$$

$$= [B(A + 1 + C) + AC] [A + C]$$

$$= [B + AC] [A + C] = AB + BC + CA$$

$$f^c = (\bar{A} + \bar{B}) \cdot (\bar{B} + \bar{C}) \cdot (\bar{C} + \bar{A})$$

(*) n Variables $\rightarrow 2^n$ Total Possible $\rightarrow 2^{n-1}$ are
distinct
Boolean fn. Self-dual

Eg: 4 Variable $\Rightarrow 2^4 = 16$ Possible Distinct fn.

$$\Rightarrow 2^3 = 8 = 256 \text{ (Self dual fn.)}$$

END OF
DAY-5